13. Thought as Symbol Processing

Quotes: Kenneth Craik, *The Nature of Explanation* (1943), ch. 5, "Hypothesis on the Nature of Thought."

Craik's Hypothesis on the Nature of Cognition

Writing in 1943, on the cusp of what would soon be recognized as the "cognitive revolution," Craik envisioned a mechanistic theory of the mind:

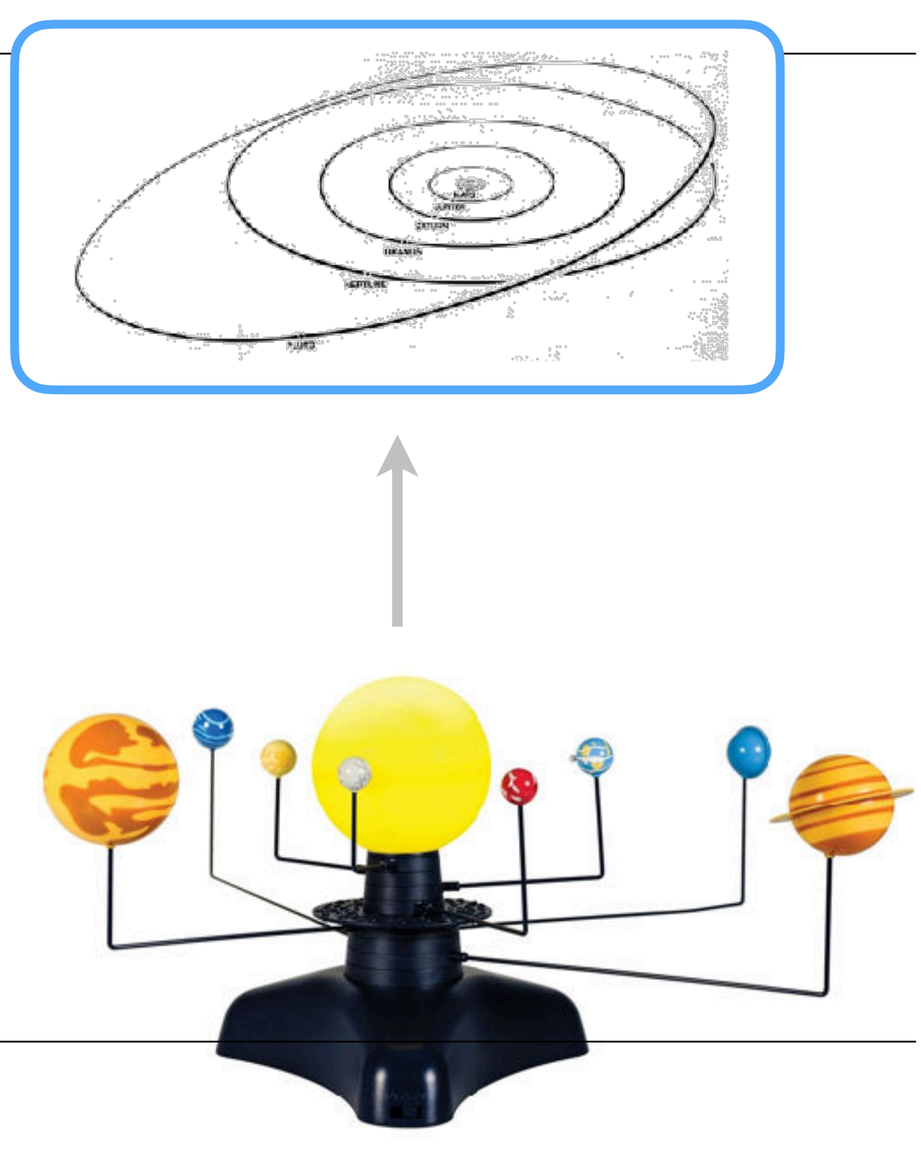
> My hypothesis then is that thought models, or parallels, reality— that its essential feature is not 'the mind', 'the self', 'sense-data', nor propositions but symbolism, and that this symbolism is largely of the same kind as that which is familiar to us in mechanical devices which aid thought and calculation. (57)

There are two parts to this hypothesis: (i) that "thought models, or parallels, reality"; and (ii) that this modeling is achieved through "symbolism" analogous to that of calculating machines.

Physical Models of Reality

Let's start with the observation that we often use physical models to help understand other phenomena. Like words or pictures these models represent external events. But they do more than that. They change and evolve to match the way the represented system changes and evolves.

> The physical process which it is desired to predict is *imitated* by some mechanical device or model which is cheaper, or quicker, or more convenient in operation. (51)



Thought as a Model of Reality

Craik proposes that thought may have a similar function of modeling external processes.

> Human thought has a definite function; it provides a convenient small-scale model of a process so that we can, for instance, design a bridge in our minds and know that it will bear a train passing over it instead of having to conduct a number of full-scale experiments; and the thinking of animals represents on a more restricted scale the ability to represent, say, danger before it comes and leads to avoidance instead of repeated bitter experience. (59)

But he notes that this can't be achieved by actually *duplicating* the external object within.

> This process of reasoning has produced a final result similar to that which might have been reached by causing the actual physical processes to occur (e.g. building the bridge haphazard and measuring its strength or compounding certain chemicals and seeing what happened); but it is also clear that this is not what has happened; the man's mind does not contain a material bridge or the required chemicals. (51)

Symbolic Models of Reality

The key insight is to realize that modeling can be achieved by "relation-structures" involving symbols.

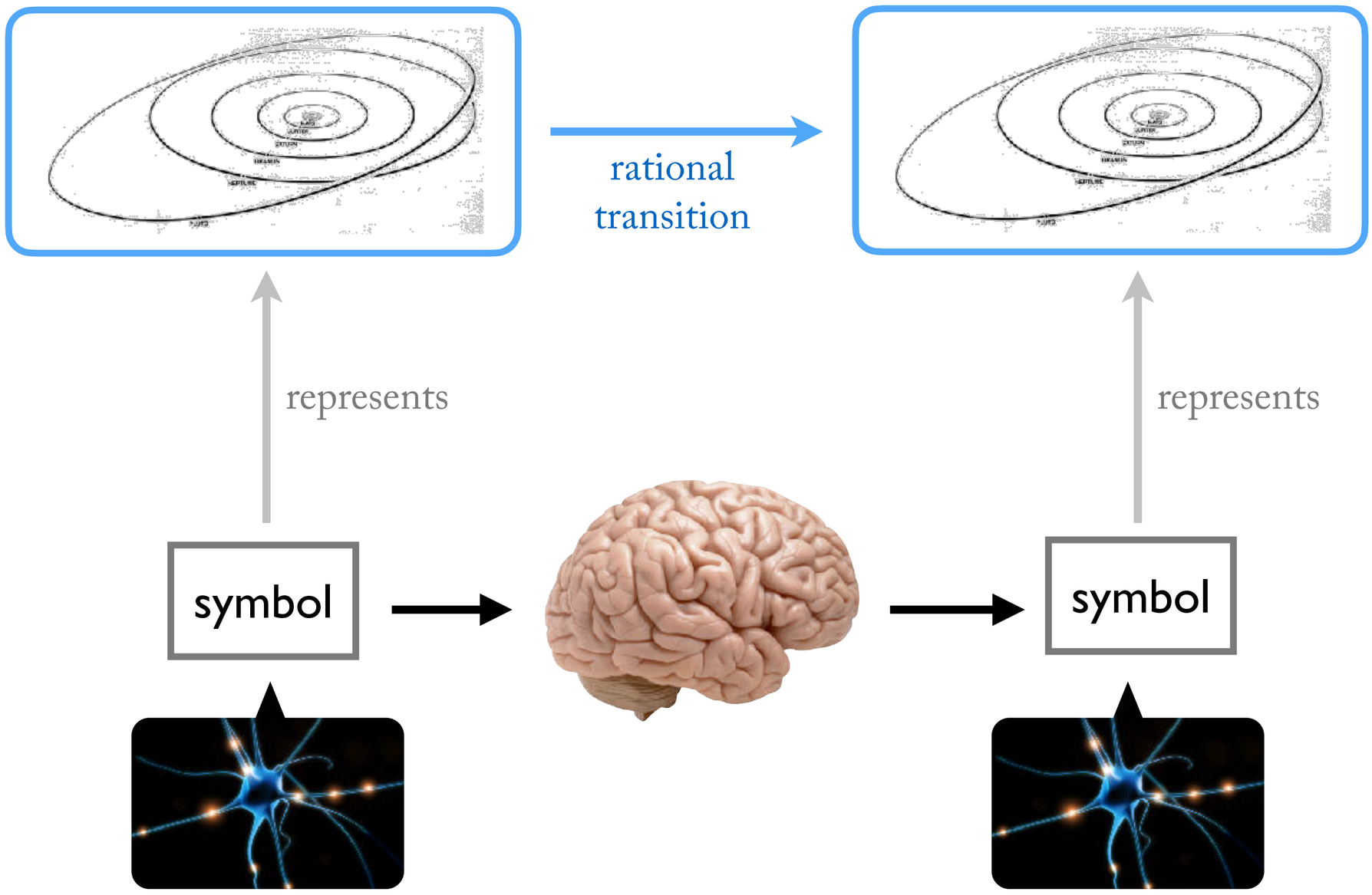
> By a model we thus mean any physical or chemical system which has a similar relation-structure to that of the process it imitates. By 'relation-structure' I do not mean some obscure non-physical entity which attends the model, but the fact that it is a physical working model which works in the same way as the process it parallels, in the aspects under consideration at any moment. Thus, the model need not resemble the real object pictorially. (51)

This leads to Craik's core hypothesis:

> The process of thought, reduced to its simplest terms, is as follows…
> (1) '**Translation**' of external process into words, numbers or other symbols,
> (2) Arrival at other symbols by a process of '**reasoning**', deduction, inference, etc., and
> (3) '**Retranslation**' of these symbols into external processes (as in building a bridge to a design) or at least recognition of the correspondence between these symbols and external events (as in realising that a prediction is fulfilled). (50; emphasis added)

And the central analogy between minds and computers:

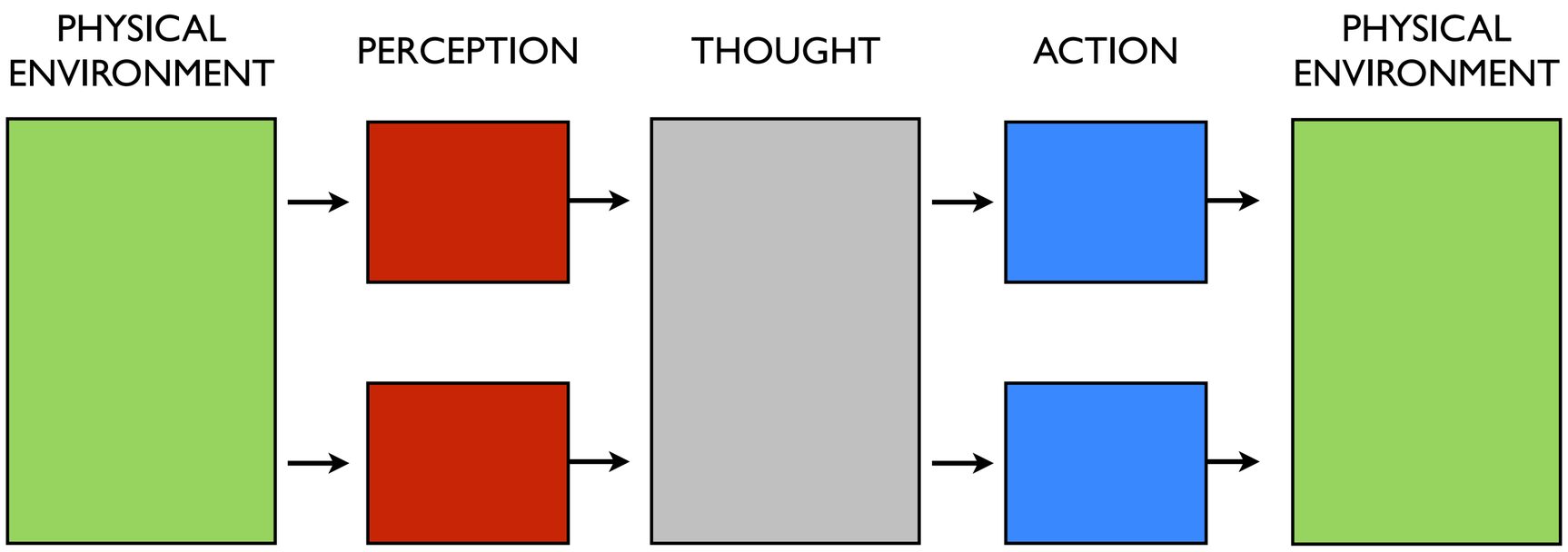
> I have tried… to indicate what I suspect to be the fundamental feature of neural machinery— its power to parallel or model external events— and have emphasised the fundamental role of this process of paralleling in calculating machines. (52)

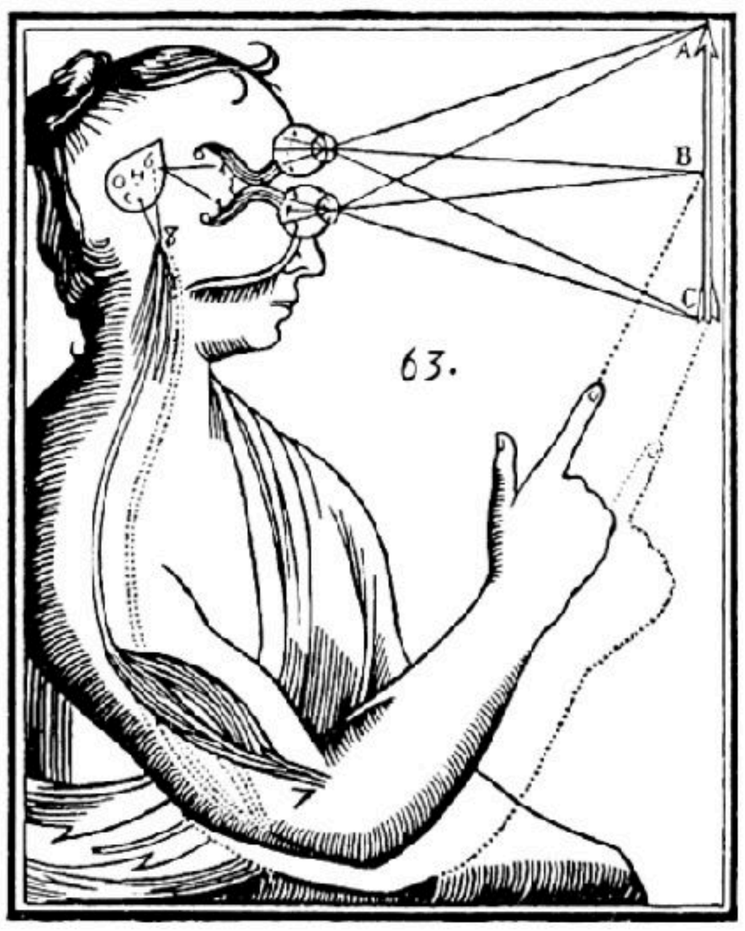


Mind as Input/Output Machine

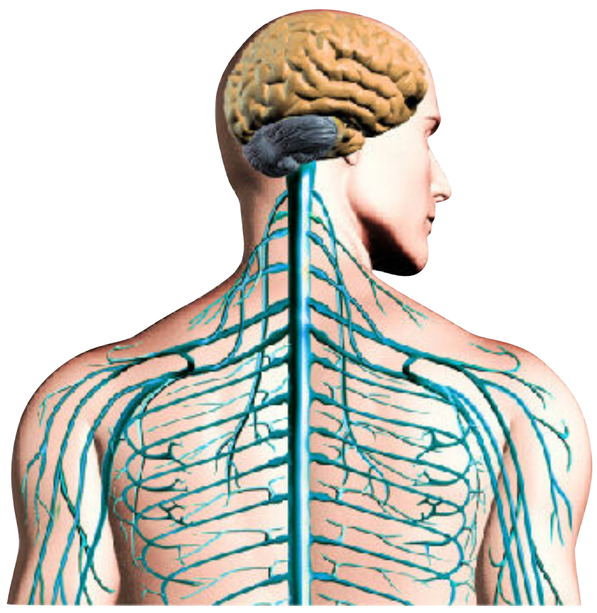
> …it does not seem overbold to consider whether the brain does not work in this way—that it imitates or models external processes. The three processes of translation, inference, and retranslation then become the translation of external events into some kind of neural patterns by stimulation of the sense-organs, the interaction and stimulation of other neural patterns as in 'association', and the excitation by these of effectors or motor organs. (53)

This three-part division of cognition is as old as Descartes, and it is the foundation of contemporary cognitive science:





For Descartes, the mind was made up of the *passions*, *thinking*, and *actions* of an immaterial soul.



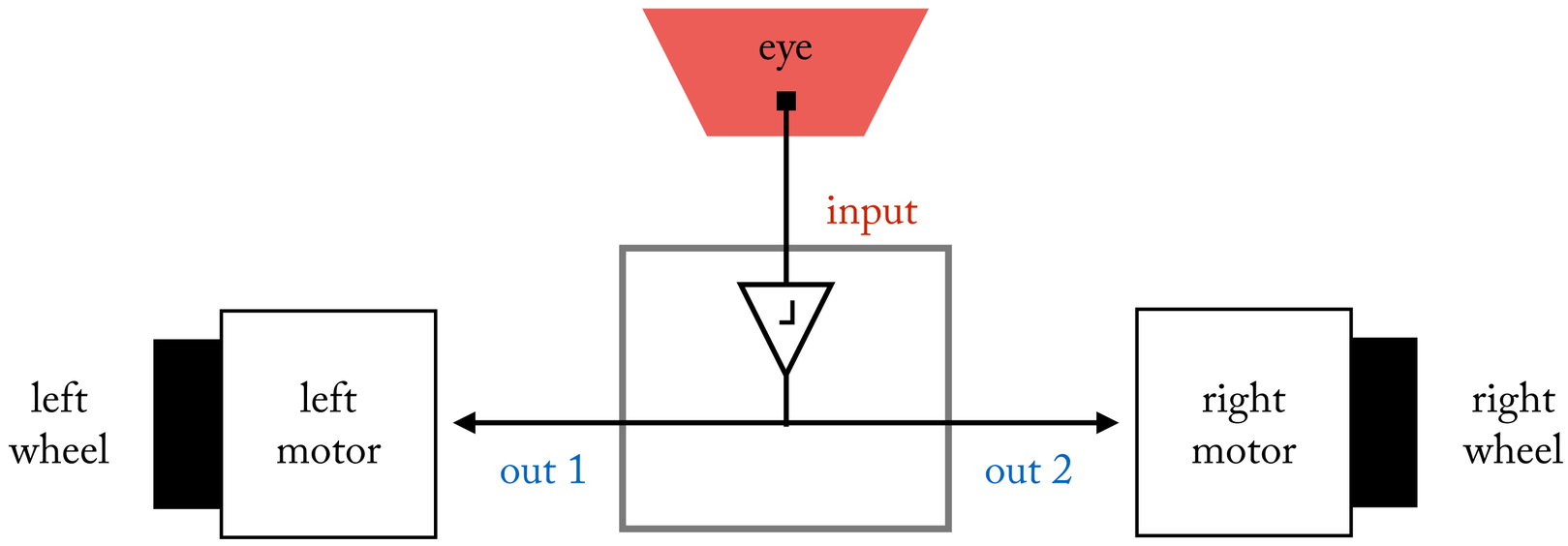
Neuroscience recognizes a central nervous system with inputs from the sensory systems and outputs to the motor systems.



Even in the humble turbot, simple perception and action abilities are mediated by logic circuits.

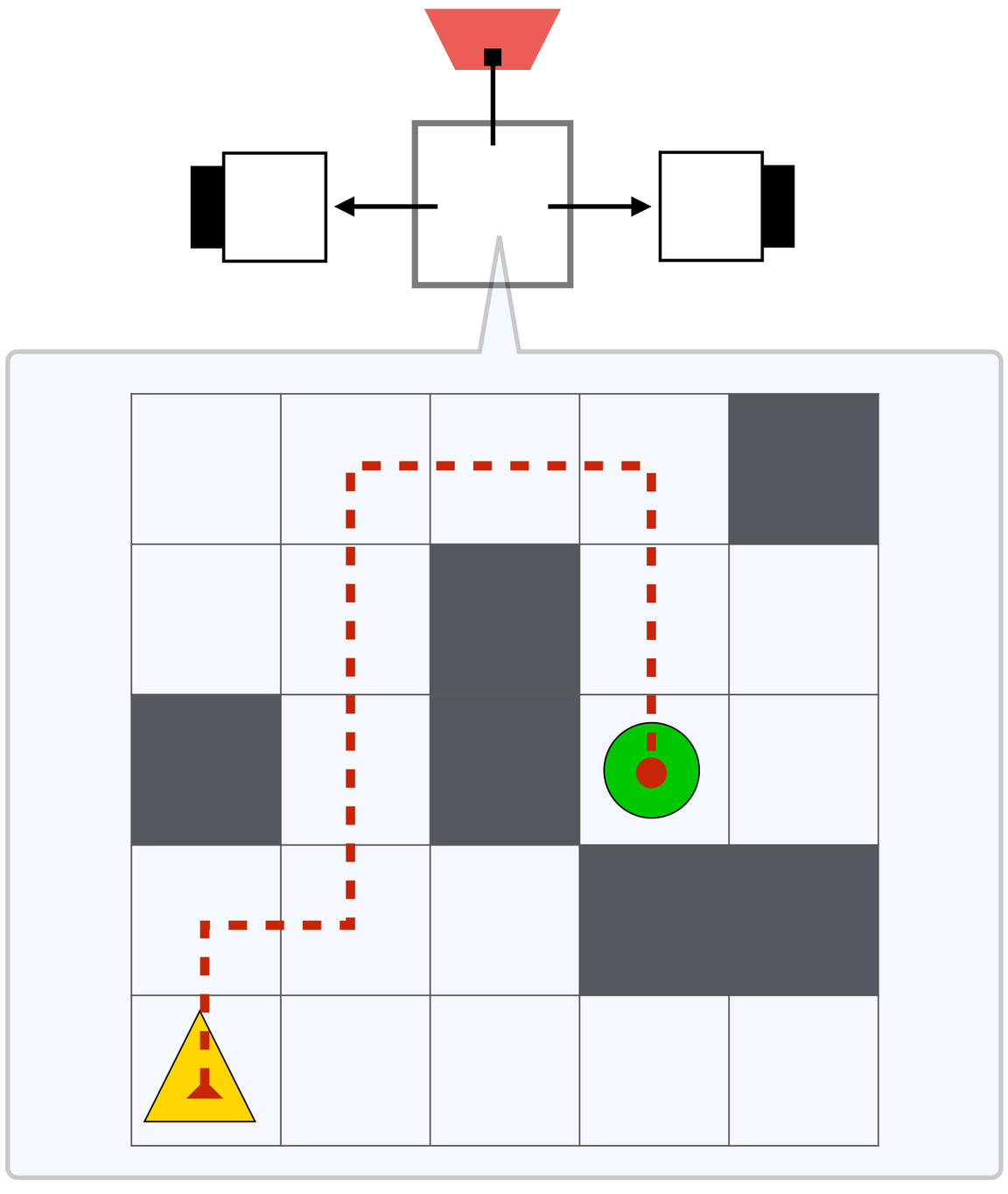
Turbots and Symbolic Modeling

By design, turbots contain the three key elements of cognition that Craik identified: (1) a capacity to perceive the external world; (2) a capacity to symbolically manipulate the inputs from perception; (3) a capacity to turn the outputs of this process into actions.



But so far, the turbots we are considering don't have anything like an "internal model". They *just react* to the environment, where their reactions are determined directly by their input and by their circuitry, but not by any internal information.

But there is no in-principle obstacle to constructing more complex turbots, ones that can learn from the environment, create rich internal models, and work out solutions to problems of navigation before physically pursuing them.



If the organism carries a 'small-scale model' of external reality and of its own possible actions within its head, it is able to try out various alternatives, conclude which is the best of them, react to future situations before they arise, utilise the knowledge of past events in dealing with the present and future, and in every way to react in a much fuller, safer, and more competent manner to the emergencies which face it. (61)